

THE CIRCLE IS NOT THE POISSON BOUNDARY OF THOMPSON'S GROUP T

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ABSTRACT.

1. INTRODUCTION

Let G be a group of orientation-preserving homeomorphisms of the circle S^1 that acts proximally on S^1 , that is, such that for all $x \neq y \in S^1$ and $\varepsilon > 0$ there is a $g \in G$ with $d(gx, gy) < \varepsilon$. Given a nondegenerate measure μ on G , by [DKN07] there exists a unique μ -stationary probability measure ν on S^1 , and moreover for almost every realization of the left random walk $\mathbf{w} = \{w_n\}_{n \in \mathbb{N}} \in G^{\mathbb{N}}$ driven by μ there exists a point $\xi(\mathbf{w}) \in S^1$ such that

$$w_n \nu \xrightarrow[n \rightarrow \infty]{} \delta_{\xi(\mathbf{w})}$$

in the weak-* topology. The distribution of $\xi(\mathbf{w})$ coincides with ν , and furnishes a natural example of a (G, μ) -boundary in the sense of Furstenberg.

By work of B. Deroin [Der13], this specific boundary is known to coincide with the whole Poisson boundary of (G, μ) for a class of groups G including cocompact lattices in $\mathrm{PSL}_2(\mathbb{R})$. Nevertheless, this class is confined to groups satisfying very stringent global discreteness properties, analogous to those of (locally discrete) analytic homeomorphisms of S^1 .¹

In this note we are interested in Thompson's group T , which is the group of dyadic piecewise affine homeomorphisms of the circle: this is the group of orientation-preserving homeomorphisms g of S^1 such that the derivative of g is defined outside a finite set inside $\mathbb{Z}[1/2]/\mathbb{Z}$ and takes values in $\{2^k\}_{k \in \mathbb{Z}}$. The objective of this paper is to prove that (S^1, ν) is not the Poisson boundary of the random walk on T driven by μ . This answers a question asked in [Der13, Section 6], and which is asked again in [Nav18, Question 19] for finitely supported, non-degenerate and symmetric probability measures on T .

We answer this question in two ways. The first one consists in exhibiting a harmonic function that does not arise from the boundary (S^1, ν) . This harmonic function comes from a boundary originally constructed by V. Kaimanovich for (F, η) in [Kai17], where F is Thompson's group acting on $[0, 1]$ and η is a nondegenerate finitely supported measure. To construct this boundary $(X, \tilde{\nu})$ on (T, μ) , let $X = \mathrm{conf}(\mathbb{Z}[1/2]/\mathbb{Z}, \mathbb{Z})$ be the set of functions from $\mathbb{Z}[1/2]/\mathbb{Z}$ to $\mathbb{Z}$. The vertex set of the Schreier graph of the T -orbit of 0 on S^1 is $\mathbb{Z}[1/2]/\mathbb{Z}$, and the random walk driven by μ on this

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¹When G is composed of analytic maps, these conditions conjecturally imply that G is virtually free or conjugate by an analytic diffeomorphism to a cocompact lattice of $\mathrm{PSL}_2(\mathbb{R})$, see [Der20].

graph is transient. Much like on lamplighter groups $\mathbb{Z}/2\mathbb{Z} \wr \mathbb{Z}$, the logarithms of the discontinuities of the derivative of the random walk on T stabilize towards a random function $\mathcal{C}_\infty \in X$. We denote the hitting measure of $\mathcal{C}_\infty$ by $\tilde{\nu}$, which will be nonatomic.

Proposition A. Suppose that μ is a nondegenerate measure on T with finite first moment. Then the boundary $(X, \tilde{\nu})$ is not a G -equivariant factor of (S^1, ν) .

Apart from other microsupported groups of piecewise affine or piecewise projective homeomorphisms of S^1 , the proof of Proposition A does not generalize well beyond this piecewise setting.

The second answer consists in an adaptation to the conditional entropy setting of a method due to A. Erschler [Ers04], which was used to show that Poisson boundaries of nontrivial wreath products are nontrivial whenever μ has finite entropy and the projection to the base group is transient. This method generalizes well to groups of diffeomorphisms of S^1 acting proximally that contain a nontrivial element whose support is a strict subinterval of S^1 .

We need some notation: the *support* of an element $a \in \text{Homeo}(S^1)$ is $\text{supp}(a) = \overline{\{x \in S^1 \mid a(x) \neq x\}}$. For $\tau \in (0, 1)$ denote $\text{Diff}_+^{1+\tau}(S^1)$ for the group of orientation-preserving diffeomorphisms of S^1 with τ -Hölder derivative. For a function $\phi: S^1 \rightarrow \mathbb{R}$, write

$$|\phi|_{\text{Lip}} = \sup_{x \neq y \in S^1} \frac{|\phi(x) - \phi(y)|}{d(x, y)} \quad \text{and} \quad |\phi|_\tau = \sup_{x \neq y \in S^1} \frac{|\phi(x) - \phi(y)|}{d(x, y)^\tau}.$$

Theorem B. Let $G \leq \text{Diff}_+^1(S^1)$ be a group acting proximally and minimally on S^1 such that there exists an $a \in G$ with support equal to a closed proper subinterval of S^1 . Let μ be a nondegenerate measure on G such that either

- the support of μ is finite, or
- the support of μ is included in $\text{Diff}_+^{1+\tau}(S^1)$ for some $\tau \in (0, 1)$ and the integrals

$$\int_G \max \left\{ |g|_{\text{Lip}}, |g^{-1}|_{\text{Lip}} \right\}^\delta d\mu(g) \quad \text{and} \quad \int_G |\log g'|_\tau d\mu(g)$$

are finite for some $\delta > 0$.

Then (S^1, ν) is not the Poisson boundary of (G, μ) .

Notice that Theorem B generalizes Proposition A when μ is of finite support, since by [GS87, Théorème A] there exists $h \in \text{Homeo}_+(S^1)$ such that hTh^{-1} lies in $\text{Diff}_+^\infty(S^1)$.

2. PRELIMINARIES

In this section we recall the theory of random walks on groups and of groups acting on the circle. For more information on this material see [Lal23, Ghy01].

2.1. Groups acting on the circle. Recall important things about groups acting on the circle.

3. PRELIMINARIES: THE ENTROPY METHOD FOR POISSON BOUNDARIES

Recall that the *entropy* of a discrete random variable X is

$$H(X) := - \sum_x \mathbb{P}(X = x) \cdot \log P(X = x).$$

Given a σ -algebra $\mathcal{F}$, the *conditional entropy* of X is

$$H(X|\mathcal{F}) := -\mathbb{E} \left[\sum_x \mathbb{P}(X = x|\mathcal{F}) \cdot \log P(X = x|\mathcal{F}) \right].$$

Recall that we always have $H(X) \leq H(X|\mathcal{F})$. Also, for any discrete random variables X and Y $H((X, Y)) \leq H(X) + H(Y)$ with equality if and only if X and Y are independent.

Consider the Borel σ -algebra of $\mathbb{S}^1$ which gives us a σ -algebra of tail events of $G^{\mathbb{N}}$ by considering the events $\{\xi(w) \in B\}$ for every Borel subset $B \subset \mathbb{S}^1$. For any discrete random variable on $G^{\mathbb{N}}$, we will write $H(X|\mathbb{S}^1) = -\mathbb{E}_{\xi} H^{\xi}(X)$ where $H^{\xi_0}(X) = H(\{X|\xi(X) = \xi_0\})$.

3.1. Kaimanovich's relative entropy criterion. The following criterion goes back to [Kai85, Theorem 2]. See also [Kai00, Theorem 4.6] or Theorem 14.35 in the book of Lyons-Peres. Recall that a μ -boundary (X, λ) of a countable group G with probability measure μ on G is

Theorem 3.1 (Kaimanovich's conditional entropy criterion). *Let G be a countable group and let μ be a probability measure on G with $H(\mu) < \infty$. Consider a μ -boundary (X, λ) of G . Then (X, λ) is the Poisson boundary of (G, μ) if and only if*

$$\lim_{n \rightarrow \infty} \frac{H(w_n|X)}{n} = 0$$

where $\{w_n\}_n$ is the μ -random walk on G .

To prove Theorem B, it suffices to show

$$H(w_n|\mathbb{S}^1) \gtrsim n.$$

3.2. Erschler's lower bound on entropy. To show a lower bound on the conditional entropy, we will use a method of [Ers04]. In [Ers04], the method is used for unconditional entropy, to show that certain groups are non-Louville. Here we will use this method for the conditional entropy.

WRITING QUESTION: Maybe add 3 sentences about 3D lamplighters for context?

Next, we recall some definitions regarding Erschler's method and state the unconditional version of the theorem.

Let G be a countable group and μ is a probability measure on G with $H(\mu) < \infty$. Consider a non-trivial μ -boundary (X, λ) of G .

A *collection of length $n \in \mathbb{N}$* is a tuple $(w, i_1, \dots, i_k)$ where $0 < i_1 < \dots < i_k \leq n$ are integers and

$$w = (x_1, x_2, \dots, x_{i_1-1}, x_{i_1+1}, \dots, x_{i_2-1}, x_{i_2+1}, \dots, x_{i_k-1}, x_{i_k+1}, \dots, x_n)$$

is an $(n - k)$ -tuple of elements of $\text{supp}(\mu)$. Notice that we index the elements of w by integers in $\{1, \dots, n\} \setminus \{i_1, \dots, i_k\}$.

WRITING CONCERN: We really need a name for the parameter k . It sounds too technical to refer to " k " all the time.

Given $a, b \in \text{supp}(\mu)$ and $q = (w, i_1, \dots, i_k)$ a collection of length n , define $T^{a,b}(q)$ as the *set of trajectories* $(y_1, \dots, y_n) \in G^n$ such that (after setting $x_0 = e_G$)

- for all $l \in \{1, \dots, n\} \setminus \{i_1, \dots, i_k\}$ we have $y_l = y_{l-1}x_l$, and
- for all $l \in \{i_1, \dots, i_k\}$, we have $y_l = y_{l-1}a$ or $y_l = y_{l-1}b$.

Thus $T^{a,b}(q)$ is a set of 2^k trajectories. We say that $T^{a,b}(q)$ is *satisfactory* if all trajectories in $T^{a,b}(q)$ arrive at different elements of G at time n .

Theorem 3.2 (Erschler [Ers04]). *Suppose that there exist $p, c > 0$ such that for each $n \in \mathbb{N}$ there is a set A_n of collections of length n such that*

- (1) *for each $q = (w, i_1, \dots, i_k) \in A_n$ we have $k \geq cn$.*
- (2) *for each $q \in A_n$ the set of trajectories $T^{a,e_G}(q)$ is satisfactory.*
- (3) *for each $q_1, q_2 \in A_n$ with $q_1 \neq q_2$ we have $T^{a,e_G}(q_1) \cap T^{a,e_G}(q_2) = \emptyset$.*
- (4) *for each $n \geq 1$ it holds that*

$$\mathbb{P} \left[\bigcup_{q \in A_n} T^{a,e_G}(q) \right] \geq p.$$

Then $H(w_n) \gtrsim n$.

3.3. Using disjoint intervals to construct satisfactory sets of trajectories. In order to show $H(w_n | \mathbb{S}^1) \gtrsim n$, we will construct for each $n \in \mathbb{N}$ a set A_n of collections of length n . Fix an element $a \in G$, $a \neq 1$, and a closed interval $I \subset S^1$ such that $\text{supp}(a) \subseteq I$.

Consider a collection $(w, i_1, \dots, i_k)$ where $w = \{x_l\}_{l \in \{1, \dots, n\} \setminus \{i_1, \dots, i_k\}}$ and define for all $1 \leq j \leq k$ the elements

$$\tilde{w}_j = x_1 x_2 \cdots x_{i_1-1} x_{i_1+1} \cdots x_{i_j-1} \quad (1)$$

and

$$\tilde{g}_j = x_{i_{j-1}+1} \cdots x_{i_j-1},$$

so $\tilde{w}_j = \tilde{g}_1 \cdots \tilde{g}_j$. The product in (1) is taken over the indices $l \in \{1, \dots, n\} \setminus \{i_1, \dots, i_k\}$, $l < i_j$. By definition, we set $i_0 = 0$.

We say that $(w, i_1, \dots, i_k)$ is *good* if the intervals $I, \tilde{w}_1(I), \dots, \tilde{w}_k(I)$ are pairwise disjoint.

Proposition 3.3. *Let q be a good collection. Then $T^{a,e_T}(q)$ is satisfactory.*

Proof. Write $q = (w, i_1, \dots, i_k)$ and $\tilde{g}_1, \dots, \tilde{g}_k, \tilde{w}_1, \dots, \tilde{w}_k$ as in the definition of a good collection. Consider two distinct trajectories $\mathbf{f}, \mathbf{h} \in T^{a,e_T}(q)$ and let $1 \leq j \leq k$ be the largest index such that the increments at time i_j in $\mathbf{f}$ and $\mathbf{h}$ are different. We may assume that this increment is a in $\mathbf{f}$ and e_G in $\mathbf{h}$. Write

$$f_n = \tilde{g}_1 a_1 \tilde{g}_2 a_2 \cdots a_{j-1} \tilde{g}_j a u \quad \text{and} \quad h_n = \tilde{g}_1 b_1 \tilde{g}_2 b_2 \cdots b_{j-1} \tilde{g}_j u$$

where the a_k, b_k are in $\{a, e_G\}$ for all $k = 1, \dots, j-1$.

We claim that

$$(\tilde{g}_1 a_1 \cdots a_{j-1} \tilde{g}_j)|_I = (\tilde{g}_1 b_1 \cdots b_{j-1} \tilde{g}_j)|_I.$$

Indeed, first notice that since the intervals

$$\tilde{w}_{j-1}(I), \tilde{w}_{j-2}(I), \dots, \tilde{w}_1(I)$$

are disjoint from $\tilde{w}_j(I)$, it follows that the intervals

$$\tilde{g}_j^{-1}(I), (\tilde{g}_{j-1} \tilde{g}_j)^{-1}(I), \dots, (\tilde{g}_2 \cdots \tilde{g}_j)^{-1}(I)$$

are all disjoint from I . Now

$$(a_{j-1}\tilde{g}_j)|_I = (b_{j-1}\tilde{g}_j)|_I$$

since $\tilde{g}_j^{-1}(I)$ is disjoint from I . In the same way, we see inductively that for all $k = j, j-1, \dots, 1$, the equality

$$(a_k\tilde{g}_{k+1} \cdots a_{j-1}\tilde{g}_j)|_I = (b_k\tilde{g}_{k+1} \cdots b_{j-1}\tilde{g}_j)|_I$$

follows from the fact that $(\tilde{g}_k \cdots \tilde{g}_j)^{-1}(I)$ is disjoint from I . This proves the claim.

Thus f_n differs from h_n on $u^{-1}(I)$, and we conclude that $T^{a, e_T}(q)$ is satisfactory. $\square$

4. PROOF OF PROPOSITION A

4.1. The lamp boundary on T . Maybe state this for $\text{PPSL}_2(\mathbb{Z})$ too? In this subsection we revisit the construction of a boundary of Thompson's F by Kaimanovich in [Kai17] and adapt it to Thompson's T . Even though the arguments are almost identical to the ones in [Kai17, Sta21] we spell out the details for completeness.

Given an element $g \in T$, define a finitely supported function $\mathcal{C}_g: \mathbb{Z}[1/2]/\mathbb{Z} \rightarrow \mathbb{Z}$ by setting

$$\mathcal{C}_g(x) = \log_2 \left((g^{-1})'(x^+) \right) - \log_2 \left((g^{-1})'(x^-) \right)$$

for $x \in \mathbb{Z}[1/2]/\mathbb{Z}$, where $(g^{-1})'(x^+)$ (resp. $(g^{-1})'(x^-)$) is the left (resp. right) derivative of g^{-1} at x . That is, $\mathcal{C}_g(x)$ is the derivative jump of g^{-1} at x in base 2. This definition differs slightly from those used in [Kai17, Sta21], but this difference is necessary since we consider *right* random walks and the *left* action of T on S^1 .

Denote the set of all (not necessarily finitely supported) functions from $\mathbb{Z}[1/2]/\mathbb{Z} \rightarrow \mathbb{Z}$ by $\text{conf}(\mathbb{Z}[1/2], \mathbb{Z})$. Define a left action of T on $\text{conf}(\mathbb{Z}[1/2]/\mathbb{Z}, \mathbb{Z})$ by

$$(g, \mathcal{C}) \mapsto (S_g \mathcal{C}: x \in \mathbb{Z}[1/2]/\mathbb{Z} \mapsto \mathcal{C}(g^{-1}(x))).$$

By the chain rule, we have

$$\mathcal{C}_{gh}(x) = \log_2 \left((h^{-1})'(g^{-1}(x)^+) \right) - \log_2 \left((h^{-1})'(g^{-1}(x)^-) \right) + \log_2 \left((g^{-1})'(x^+) \right) - \log_2 \left((g^{-1})'(x^-) \right)$$

for all $g, h \in G$ and $x \in \mathbb{Z}[1/2]/\mathbb{Z}$, that is,

$$\mathcal{C}_{gh} = \mathcal{C}_g + S_g \mathcal{C}_h.$$

Define a second action of T on $\text{conf}(\mathbb{Z}[1/2]/\mathbb{Z}, \mathbb{Z})$ by $(g, \mathcal{C}) \mapsto \mathcal{C}_g + S_g \mathcal{C}$, so we have $\mathcal{C}_{gh} = g.\mathcal{C}_h$ for all $g, h \in T$. This is the action on $\text{conf}(\mathbb{Z}[1/2]/\mathbb{Z}, \mathbb{Z})$ that will define a certain nontrivial boundary for T .

Let μ be a nondegenerate probability measure on Thompson's group T . To prove transience of the random walks on T -orbits of elements of S^1 we emulate [Kai17, Theorem 25], for which we need a comparison lemma for Markov operators.

Proposition 4.1 ([BLP77]). *Let $P_1(\cdot, \cdot), P_2(\cdot, \cdot)$ be doubly stochastic kernels on a countable set X such that $P_2(\cdot, \cdot)$ is symmetric and there exists an $\varepsilon > 0$ such that*

$$P_1(x, y) \geq \varepsilon P_2(x, y)$$

for all $x, y \in X$.

Then the Markov process determined by P_1 and started at $x \in X$ is transient if the Markov process determined by P_2 and started at $x \in X$ is transient.

Lemma 4.2. *For any $x \in S^1$, the random walk driven by μ on $\text{Orb}_T(x)$ is transient.*

Proof. Fix $x \in S^1$ and find $f, g \in T$ such that there exist disjoint intervals $I_1, I_2, J_1, J_2 \subset S^1$ with $0 \notin I_1 \cup I_2 \cup J_1 \cup J_2$ and

$$f(S^1 \setminus I_2) \subseteq I_1, \quad g(S^1 \setminus J_2) \subseteq J_1.$$

By Klein's ping-pong lemma, f and g generate a free group inside T and the orbit of 0 under this free group is free. Let $\tilde{\mu}$ be the symmetric measure on $\{f, f^{-1}, g, g^{-1}\}$. The random walk driven by $\tilde{\mu}$ on $\text{Orb}_{\langle f, g \rangle}(x)$ starting at x is transient, since it is the symmetric random walk on the Cayley graph of the free group on two generators. Let $n \in \mathbb{N}$ such that the support of μ^{*n} contains f and g . Then there exists an $\varepsilon > 0$ such that $\mu \geq \varepsilon \tilde{\mu}$, so Proposition 4.1 gives that the random walk on $\text{Orb}_T(x)$ driven by μ^{*n} and started at x is transient. We conclude that the random walk driven by μ on $\text{Orb}_T(x)$ is transient too. $\square$

4.2. Harmonic functions. In this subsection, μ is a finitely supported nondegenerate probability measure on Thompson's group T . We denote by $\mathcal{C}_\infty: (G^\mathbb{N}, \mathbb{P}) \rightarrow (\text{conf}(\mathbb{Z}[1/2]/\mathbb{Z}, \mathbb{Z}), \tilde{\nu})$ the Kaimanovich boundary that arises from the stabilization of derivative jumps on the T -orbit of 0 in S^1 . We also write $\xi: (G^\mathbb{N}, \mathbb{P}) \rightarrow (S^1, \nu)$ for the boundary map that arises from the natural action of T on the circle.

Since the measure $\tilde{\nu}$ is nontrivial, there exists a $y \in \mathbb{Z}[1/2]/\mathbb{Z}$ and a $k \in \mathbb{Z}$ such that the bounded function $f: G \rightarrow [0, 1]$ defined on $g \in G$ by

$$f(g) = \mathbb{P}_g \left[\mathbf{w} \in G^\mathbb{N} \mid \mathcal{C}_\infty(\mathbf{w})(y) = k \right]$$

satisfies $f(e_T) > 0$. The function f is μ -harmonic because the set

$$\{\mathbf{w} \in G^\mathbb{N} \mid \mathcal{C}_\infty(\mathbf{w})(y) = k\}$$

is shift-invariant up to $\mathbb{P}$ -measure zero.

Lemma 4.3. *There exists a sequence $\{g_n\}_{n \in \mathbb{N}} \subset T$ where $\text{supp}(g_n)$ are closed intervals containing y and such that*

$$\text{diam}(\text{supp}(g_n)) \xrightarrow{n \rightarrow \infty} 0 \quad \text{and} \quad f(g_n) \xrightarrow{n \rightarrow \infty} 0.$$

Proof. Let $n \in \mathbb{N}$, $n \geq 2$ and define the function $g_n \in T$ as

$$g_n(x) = \begin{cases} y + 2^n(x - y) & \text{if } y \leq x < y + 2^{-2n} \\ y + 2^{-n} - 2^{-3n} + 2^{-n}(x - y) & \text{if } y + 2^{-2n} < x < y + 2^{-2n} + 2^{-n} \\ x & \text{elsewhere,} \end{cases}$$

see Figure 1.

Notice that

- $g_n(y) = y$,
- $\text{supp}(g_n)$ is a dyadic interval containing y and of length $2^{-n} + 2^{-2n}$, and
- the derivative jump of g_n at y is equal to 2^n .

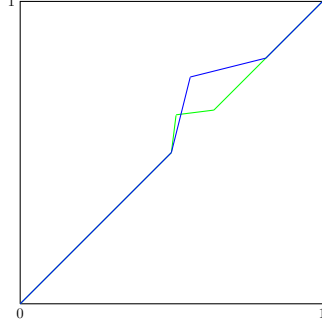


FIGURE 1. The map g_n for $y = 1/2$, $n = 2$ (blue) and $n = 3$ (green).

Thus $\text{diam}(\text{supp}(g_n)) \xrightarrow{n \rightarrow \infty} 0$. Moreover, whenever $g \in T$ fixes y we have

$$f(g) = \mathbb{P}_g [\mathcal{C}_\infty(\mathbf{w})(y) = k] = \mathbb{P} [\mathcal{C}_\infty(\mathbf{w})(y) = k - \log_2(g')^+(y) + \log_2(g')^-(y)]$$

so

$$f(g_n) = \mathbb{P} [\mathcal{C}_\infty(\mathbf{w})(y) = k - n]$$

converges to 0 as $n \rightarrow \infty$, as desired. $\square$

Proposition 4.4. *There is no function $h \in L^\infty(S^1, \nu)$ such that*

$$f(g) = \int_{S^1} h(gx) \, d\nu(x)$$

for all $g \in G$. In particular, (S^1, ν) is not the Poisson boundary of (T, μ) .

Proof. Suppose, towards a contradiction, that such a function $h \in L^\infty(S^1, \nu)$ exists. Set $I_n = \text{supp}(g_n)$ where the g_n are as in Lemma 4.3. The equality

$$f(g_n) = \int_{S^1} h(g_n x) \, d\nu(x) = \int_{S^1 \setminus I_n} h(g_n x) \, d\nu(x) + \int_{I_n} h(g_n x) \, d\nu(x)$$

and

$$\left| \int_{I_n} h(g_n x) \, d\nu(x) \right| \leq \nu(I_n) \|h\|_\infty$$

imply that

$$\int_{S^1 \setminus I_n} h(g_n x) \, d\nu(x) \xrightarrow{n \rightarrow \infty} 0,$$

since ν is nonatomic.

But

$$\int_{S^1 \setminus I_n} h(g_n x) \, d\nu(x) = \int_{S^1 \setminus I_n} h(x) \, d\nu(x),$$

hence the dominated convergence theorem shows that

$$f(e_T) = \int_{S^1} h(x) \, d\nu(x) = 0.$$

This is a contradiction. $\square$

5. THERE IS A LINEAR NUMBER OF DISJOINT INTERVALS ALONG THE WALK

NOTE TO THE AUTHORS The definition of Z_N has been slightly altered to make the notation in the entropy lower bound cleaner... (i encountered annoying technical issues when trying to properly define the set of collections and their corresponding trajectories).

Consider the space (S^1, ν) where ν is the unique μ -stationary probability measure on S^1 . From now on, fix an element $a \in T \setminus \{e_T\}$ and a closed interval $I \subset S^1$ such that $\text{supp}(a) \subseteq I$ and $\nu(I) < 1$. For every $n \in \mathbb{N}$, define the random variable $Z_n \in \mathbb{N}$ as the maximal number of mutually disjoint intervals in $\{I, w_1^{-1}(I), \dots, w_n^{-1}(I)\}$ for which the next step the random walk takes is either 1 or a :

$$Z_n := \max\{k | \exists \text{ distinct } i_1, \dots, i_k \in [n] \text{ s.t. } \{w_{i_j}(I)\}_{j=1}^k \text{ are disjoint and } \forall j \ w_{i_{j+1}} w_{i_j}^{-1} \in \{1, a\}\}.$$

Proposition 5.1 (There is a linear number of disjoint intervals along the walk).

$$\mathbb{E}[Z_N] \gtrsim n$$

Lemma 5.2 (Greedy algorithm for getting disjoint intervals). *For any $s \in \mathbb{N}$ and sufficiently large $N \in \mathbb{N}$ we have*

$$\mathbb{E}[Z_N] \geq \frac{N}{2s} (\mu(1) + \mu(a)) \mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N}].$$

Just for the sake of naming things, let us call $\mathbb{P}(w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N})$ the **probability of never intersecting with sparsity parameter s** .

Proof. We greedily choose points along the arithmetic progression $\{0, s, 2s, 3s, 4s, \dots\} \subset \mathbb{N}$. Without loss of generality let $N = ns$ for some $n \in \mathbb{N}$. For each $j = 1, \dots, n$ define the event

$$C_j = \{w_{is}(I) \cap w_{js}(I) = \emptyset \text{ for all } i = 1, \dots, j-1\}.$$

By greedy choice, the Markov property, and the reversibility of the random walk we have

$$\mathbb{E}[Z_N] \geq \sum_{j=1}^n \mathbb{P}[C_j \text{ and } w_{js+1} w_{js}^{-1} \in \{1, a\}] = (\mu(1) + \mu(a)) \sum_{j=1}^n \mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i = 1, \dots, j-1].$$

Since the events $\{w_{is}(I) \cap I = \emptyset \text{ for all } i = 1, \dots, j-1\}$ are nested in j we obtain

$$\mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i = 1, \dots, j-1] \xrightarrow{j \rightarrow \infty} \mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N}],$$

and for the Cesaro averages we conclude that

$$\frac{s}{N} \sum_{j=1}^n \mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i = 1, \dots, j] \xrightarrow{n \rightarrow \infty} \mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N}].$$

The result follows for sufficiently large N . □

Proposition 5.3 (Exponential contraction in mean to the boundary point). *There exists $\lambda > 0$ and $C > 0$ such that*

$$\sup_{x \in \mathbb{S}^1} \mathbb{E}[d(w_n x, \xi)] \leq C e^{-\lambda n},$$

Proof. See Proposition 4.4 in the first arxiv version of "Probabilistic Tits Alternative for circle diffeomorphisms" by Martin Gilabert Vio. □

Lemma 5.4 (Lower bound on the probability of never intersecting). *There exists a sparsity $s \in \mathbb{N}$ and $\Xi \subseteq \mathbb{S} \setminus I$ of measure $\nu(\Xi) \geq (1 - \nu(I))/2$ such that*

$$\mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N} \mid \xi \in \Xi] > 4/5.$$

As a consequence, we have that

$$\mathbb{E}[Z_N] \geq \frac{N}{2s}(\mu(1) + \mu(a))\nu(\Xi)\mathbb{P}[w_{is}(I) \cap I = \emptyset \text{ for all } i \in \mathbb{N} \mid \xi \in \Xi] \gtrsim N.$$

Proof. Since ν is continuous we have

$$\nu(\{\xi \mid 0 < d(\xi, I) < \varepsilon\}) \xrightarrow{\varepsilon \rightarrow 0} 0,$$

so there exists $\varepsilon > 0$ such that

$$\Xi := \{\xi \mid d(\xi, I) > \varepsilon\}$$

has ν -measure $\geq (1 - \nu(I))/2$. Now choose a sparsity s such that $e^{-\lambda s/2} < \min\{\varepsilon\nu(\Xi)/2, 1/10\}$, and denote the endpoints of the interval by $I = [l, r]$. For each $i \in \mathbb{N}$ we use the assumption and the Markov inequality to get

$$\begin{aligned} \nu(\Xi)\mathbb{P}[w_{is}(I) \cap I \neq \emptyset \mid \xi \in \Xi] &\leq \nu(\Xi)\mathbb{P}[\max\{d(w_{is}(l), \xi), d(w_{is}(r), \xi)\} > \varepsilon \mid \xi \in \Xi] \\ &= \mathbb{P}[\max\{d(w_{is}(l), \xi), d(w_{is}(r), \xi)\} > \varepsilon] \leq e^{-\lambda is} \frac{2}{\varepsilon} \end{aligned}$$

and hence

$$\mathbb{P}[w_{is}(I) \cap I \neq \emptyset \mid \xi \in \Xi] \leq 2e^{-\lambda s/2}e^{-\lambda(i-1)s}.$$

By the union bound we get a geometric series and conclude that

$$\mathbb{P}[\text{there exists } i \text{ such that } w_{is}(I) \cap I \neq \emptyset \mid \xi \in \Xi] \leq 2e^{-\lambda s/2} \leq 1/5. \quad \square$$

6. AT MANY TIMES, THE CONDITIONAL TRANSITION PROBABILITIES DO NOT CHANGE

For λ -almost every $\xi \in X$ one can consider the conditional probability measure $\mathbb{P}^\xi$, so that one has $\mathbb{P} = \int_X \mathbb{P}^\xi d\lambda(\xi)$. These measures are Markov and determine homogeneous (in time, not space) Markov chains on G with transition probabilities

$$\mathbb{P}^\xi[w_{n+1} = g \mid w_n = h] = \mathbb{P}[w_{n+1} = g \mid w_n = h] \frac{dg_*\lambda}{dh_*\lambda}(\xi) = \mu(h^{-1}g) \frac{dg_*\lambda}{dh_*\lambda}(\xi),$$

for each $g, h \in G$. (This is easy to see. For a reference, look at Lemma 14.33 on page 490 in the book of Lyons-Peres) The following Lemma implies that the probability of multiplying by $a \in G$ in the conditional random walk is $\mu(a)$ whenever $w_n = g$ and the limit point $\xi \notin gI$.

Lemma 6.1. *For every $g \in G$ and λ -almost every $\xi \notin gI$ we have*

$$\frac{d(ga)_*\lambda}{dg_*\lambda}(\xi) = 1.$$

Proof. Indeed, let $A \subseteq \mathbb{S}^1 \setminus g(I)$ be any measurable subset with $\lambda(A) > 0$ and $\xi \in A$. Then

$$g_*\lambda(A) = \lambda(g^{-1}A) = \mathbb{P}[\xi(\mathbf{w}) \in g^{-1}A] = \mathbb{P}[\xi(\mathbf{w}) \in (ga)^{-1}A] = (ga)_*\lambda(A).$$

The third step follows from the fact that $a^{-1}g^{-1}A = g^{-1}A$ which follows from $g^{-1}A \subset \mathbb{S}^1 \setminus I$, that is, $A \subset \mathbb{S}^1 \setminus g(I)$. $\square$

Next, we wish to estimate the number of times n for which $w_n^{-1}\xi \notin I$.

Proposition 6.2 (Equidistribution). *For ν -almost every $\xi \in S^1$, for $\mathbb{P}^\xi$ -almost every $\mathbf{w} = \{w_n\}_{n \in \mathbb{N}} \in G^\mathbb{N}$ the convergence of probability measures*

$$\frac{1}{n} \sum_{k=0}^n 1_{w_n^{-1}(\xi)} \xrightarrow{n \rightarrow \infty} \nu$$

holds in the weak- topology.*

Proof. [Mal17, Proposition 4.9] shows that for any $x \in S^1$, $\mathbb{P}$ -almost surely the convergence

$$\frac{1}{n} \sum_{k=0}^n 1_{w_n^{-1}(x)} \xrightarrow{n \rightarrow \infty} \nu \quad (2)$$

holds in the weak-* topology. We conclude that for ν -almost all ξ , (2) holds $\mathbb{P}^\xi$ -almost surely.

Fix an arbitrary $x \in S^1$. Recall that we denote $\xi: (G^\mathbb{N}, \mathbb{P}) \rightarrow (S^1, \nu)$ the boundary map that arises from the action of G on the circle.

Theorem ?? shows that there exists $\lambda > 0$ such that

$$\mathbb{E} \left[d \left(w_n^{-1}(x), w_n^{-1}(\xi(\mathbf{w})) \right) \right] \leq e^{-\lambda n}$$

for all $n \in \mathbb{N}$, so in particular the sum

$$\sum_{n \geq 0} d \left(w_n^{-1}(x), w_n^{-1}(\xi(\mathbf{w})) \right)$$

is finite $\mathbb{P}$ -almost surely. Thus for ν -almost every ξ , the sum

$$\sum_{n \geq 0} d(w_n^{-1}(x), w_n^{-1}(\xi))$$

is finite $\mathbb{P}^\xi$ -almost surely. This condition coupled with (2) gives the conclusion. $\square$

Corollary 6.3. *For ν -almost every $\xi \in S^1 \setminus I$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$,*

$$\mathbb{E}^\xi [|0 \leq k \leq n \mid \xi \in w_k I|] \leq 2\nu(I)n.$$

Proof. The statement follows from the equality

$$\mathbb{E}^\xi [|0 \leq k \leq n \mid \xi \in w_k I|] = \sum_{k=0}^n \mathbb{P}^\xi [w_n^{-1}(\xi) \in I]$$

and the equidistribution of $\{w_n^{-1}(\xi)\}_{n \in \mathbb{N}}$ from Proposition 6.2. $\square$

Conclude that for a dense subset of the times, the conditional transition probabilities are identical to the unconditional ones.

7. THE LOWER BOUND ON THE RELATIVE ENTROPY AND PROOF OF THEOREM B

QUESTION: So we pick up an element a whose support I has arbitrarily small measure $\nu(I)$, and find n_0 sufficiently high such that $\mu^{*n_0}(a) > 0$. Why do we know such an a exists?? Is this some structural result about proximal subgroups?

We saw that there exists $0 < c < 1$ such that $\mathbb{E}Z_n \geq cn$. Pick a such that $\nu(I) \leq c/8$

We recall the following basic fact about random variables, which is pigeonholing in disguise. Let $0 \leq X \leq 1$ be a real-valued random variable with mean $\mathbb{E}[X] > \lambda > 0$. Then

$$\mathbb{P}(X > \lambda/2) \geq \lambda/2.$$

(Why? We have $\lambda < \mathbb{E}[X] \leq \lambda/2\mathbb{P}(X \leq \lambda/2) + 1\mathbb{P}(X > \lambda/2) \leq \lambda/2 + \mathbb{P}(X > \lambda/2)$.)

Fix $\xi \in \mathbb{S}^1$ and consider the random variable $0 < \frac{Z_n|\xi}{n} \leq 1$. We have

$$\mathbb{P}^\xi \left(\frac{Z_n}{n} > \frac{\mathbb{E}[Z_n|\xi]}{2n} \right) \geq \frac{\mathbb{E}[Z_n|\xi]}{2n}.$$

Consider the set of infinite trajectories $\mathbf{w}$ with limit point ξ and $Z_n > \mathbb{E}[Z_n|\xi]/2$. To each such trajectory $\mathbf{w}$ we fix a set $I = \{i_1, \dots, i_k\} \subset [n]$ of $k = \lceil \mathbb{E}[Z_n|\xi]/2 \rceil$ -many times such that the intervals $\{w_{i_1}I, \dots, w_{i_k}I\}$ are disjoint and such that $w_{i_j+1}w_{i_j}^{-1} \in \{1, a\}$ for each $j = 1, \dots, k$. We assign this trajectory to the collection q consisting of times $i_1 + 1, \dots, i_k + 1$ along with all the jumps in the remaining $n - k$ times. Denote the set of all such collections by $A_{n,\xi}$. By construction,

$$\mathbb{P}^\xi \left[\bigcup_{q \in A_{n,\xi}} T^{a,1}(q) \right] \geq \frac{\mathbb{E}[Z_n|\xi]}{2n}.$$

Also, as noted in the preliminary section, each $T^{a,1}(q)$ is satisfactory (that is, any two distinct trajectories in $T^{a,1}(q)$ end up at two distinct elements at time n).

Because we chose a such that $\nu(I)$ is sufficiently small, we know that the set

$$\Xi := \left\{ \xi \in \mathbb{S}^1 \mid \frac{\mathbb{E}[Z_n|\xi]}{n} \geq 2\nu(I) \right\}$$

has ν -measure $> c/2$.

From the last section, it follows that when $\xi \in \Xi$, and $q \in A_{n,\xi}$, then for at least $k/2$ of the indices $i_1, \dots, i_k$ of the collection q we have $\xi \notin w_{i_j}I$. This means that

$$\mathbb{P}^\xi[w_{i_j+1} = aw_{i_j}] = \mu(a) \text{ and } \mathbb{P}^\xi[w_{i_j+1} = w_{i_j}] = \mu(1).$$

Write ρ for the conditional measure of μ on $\{1, a\}$. So for every $\xi \in \Xi$ and every $q \in A_{n,\xi}$, the normalized restriction of $\mathbb{P}^\xi$ restricted to $T^{1,a}(q)$ is the product measure ρ^k , so

$$H^\xi(w_n \mid w \in T^{1,a}(q)) \geq k(q) \frac{H(\rho)}{2}.$$

Thus

$$\begin{aligned}
H(w_n | \mathbb{S}^1) &= \mathbb{E}_\xi H^\xi(w_n) \geq \mathbb{E}_\xi \left(\sum_{q \in A_{n,\xi}} H^\xi(w_n | T^{a,e_G}(q)) \mathbb{P}^\xi(T^{a,e_G}(q)) \middle| \xi \in \Xi \right) \nu(\Xi) \\
&\geq \mathbb{E}_\xi \left(\left(\min_{q \in A_{n,\xi}} k(q) \right) \frac{H(\rho)}{2} \sum_{q \in A_n} \mathbb{P}^\xi(T^{a,e_G}(q)) \middle| \xi \in \Xi \right) \nu(\Xi) \\
&\geq \mathbb{E}_\xi \left(\mathbb{E}^\xi \frac{Z_n}{2} \frac{H(\rho)}{2} \mathbb{E}^\xi \frac{Z_n}{2n} \right) \frac{c}{2} \\
&\geq \frac{H(\rho)c}{16n} \mathbb{E}_\xi (\mathbb{E}^\xi Z_n)^2 \geq \frac{H(\rho)c^3}{16} n
\end{aligned}$$

where we used Jensen at the last step.

The proof of Theorem B is now complete.

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